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"EVERY GREAT SOLUTION STARTS WITH A GOOD CUP OF COFFEE."

PRESENTED BY:
NAMAN, PRIYA, VARUNA, SANJANA

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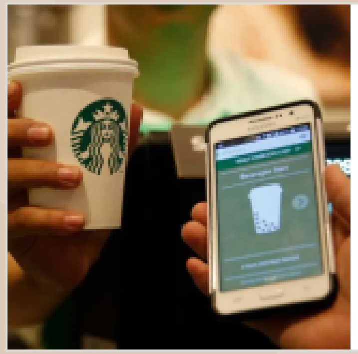
STARBUCKS DATASET



Starbucks Customer Ordering Patterns

The Digital Venti Effect: How Mobile Ordering Drives Customization and Spend.

kaggle.com

This dataset is a high - resolution behavioral map capturing the longitudinal purchasing history of 14,988 unique customers. It goes beyond simple transaction logs by tracking the relationship between specific behavioral traits such as customization frequency and visit consistency and total lifetime revenue.

Why we choose this topic

Statistical Significance, Behavioral Depth, Real-World Complexity And,We love Coffee.........

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PROBLEM DEFINITION & RESEARCH QUESTION

Central Question: What factors drive customer spending at Starbucks?

Sub-questions guiding our analysis:

1. Data Exploration (Naman): What does the spending landscape look like? Which segments show the biggest differences?
2. Correlation & Distributions (Priya): How strongly do cart size and customizations relate to spending? Does spending follow a normal distribution?
3. Hypothesis Testing (Sanjana): Are the spending differences between Mobile vs Non-Mobile and Rewards vs Non-Rewards statistically significant?
4. Probability & Bayes Theorem (Varuna): How rare are high-value spenders? Given a high-value order, who most likely placed it?

Business Motivation:

Understanding spending drivers helps Starbucks optimize its mobile app strategy, loyalty program, and in-store experience to increase revenue per order.

DATA EXPLORATION: DESCRIPTIVE STATISTICS & KEY SEGMENTS

Descriptive Statistics (Lecture 1):

total_spend:
- Mean = $14.87,
- Median = $14.17,
- SD = $5.51

Five-Number Summary:
- Min $3.51,
- Q1 $10.84,
- Median $14.17,
- Q3 $18.18,
- Max $40.31
- IQR = $7.34,

Outliers: 1,415 (1.42%) above $29.19

Mean Spend by Segment (the key finding):
- Mobile App: $18.08 (42.5% of orders)
- Drive-Thru: $12.48,
- In-Store: $12.51,
- Kiosk: $12.49

Gap: Mobile App is $5.59 above every other channel

Rewards Members vs Non-Members: $15.72 vs $14.09 (+$1.63)

Age: 18-34 spend ~$15.55, 55+ spend $13.06

Drink Category: No meaningful difference (all within $0.09)

DATA EXPLORATION: DESCRIPTIVE STATISTICS & KEY SEGMENTS

Distribution shape: Roughly symmetric with slight right skew

Mean ($14.87) close to Median ($14.17) confirms near-symmetry

Most orders fall between $10 and $20

Key EDA Takeaways:

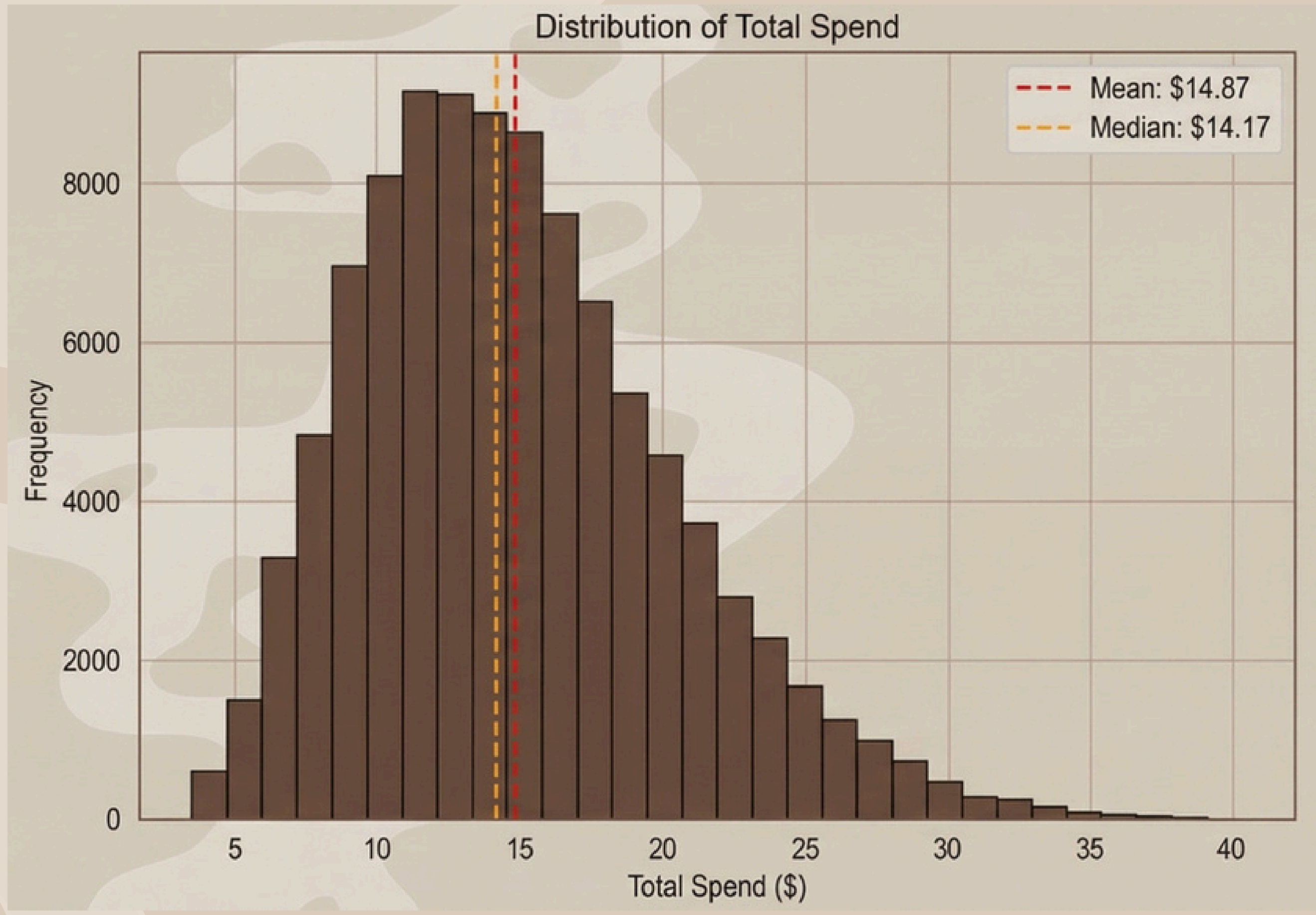
1. Mobile App is the dominant spending driver ($18.08 vs ~$12.50 for all others)

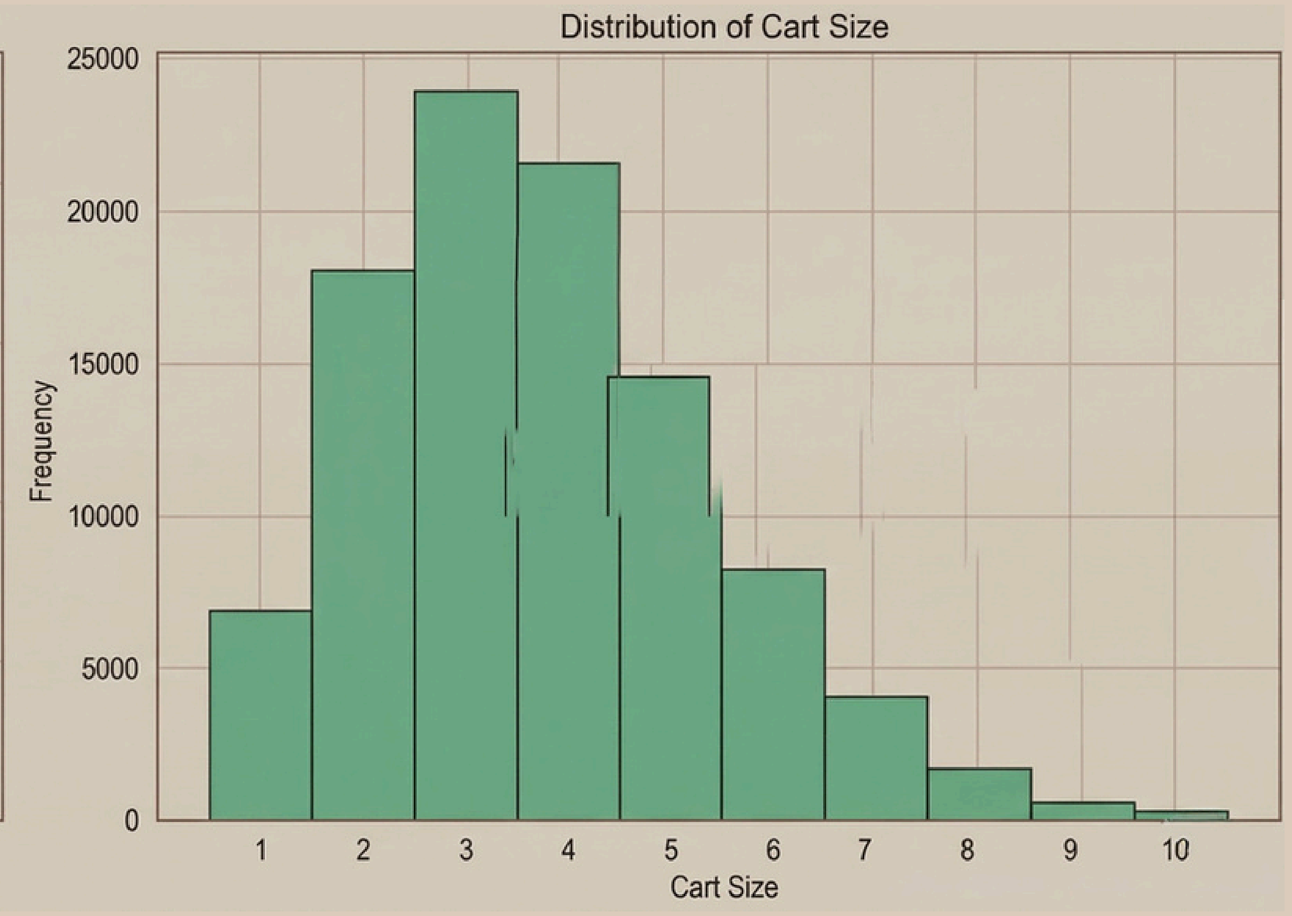
2. Rewards membership shows a modest effect (+$1.63), needs hypothesis testing

3. Drink category and store location have no meaningful effect on spending

4. Data quality is excellent: zero missing values, low outlier rates

5. These findings set the stage for all subsequent analyses





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COVARIANCE & PEARSON CORRELATION

'Covariance shows direction, while correlation shows both strength and direction on a standardized scale.'

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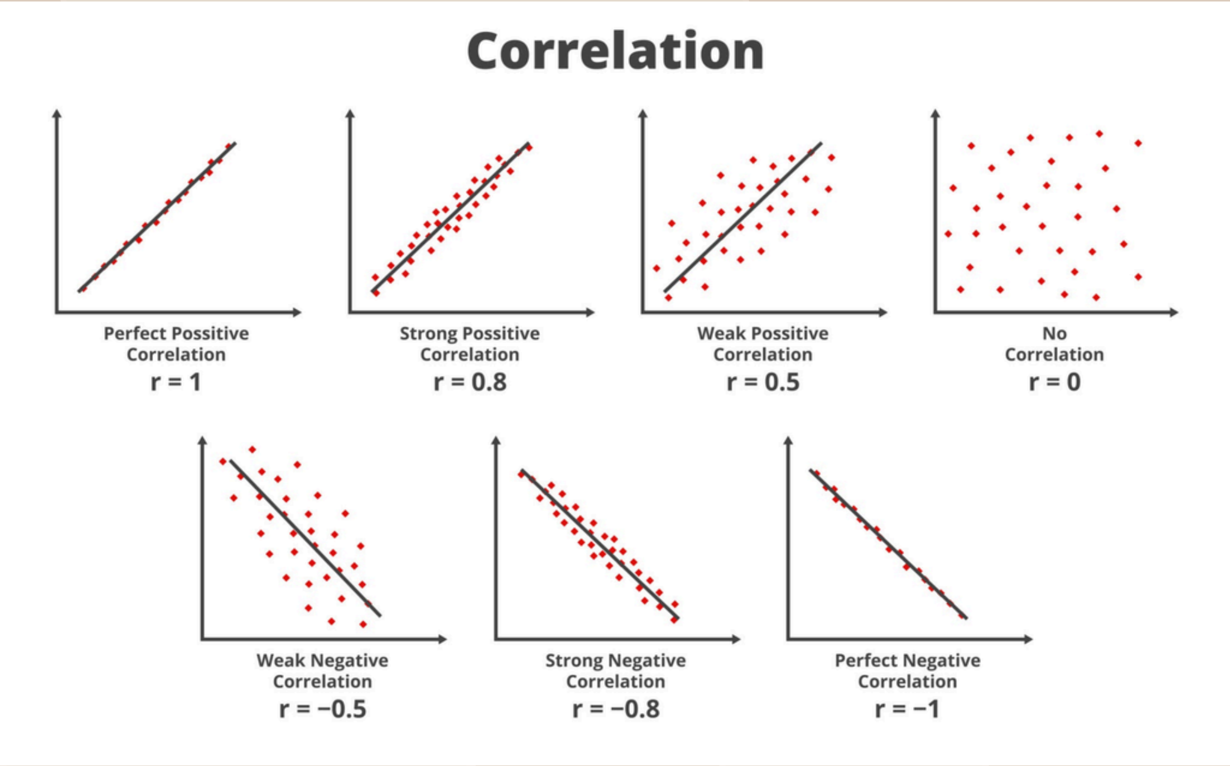
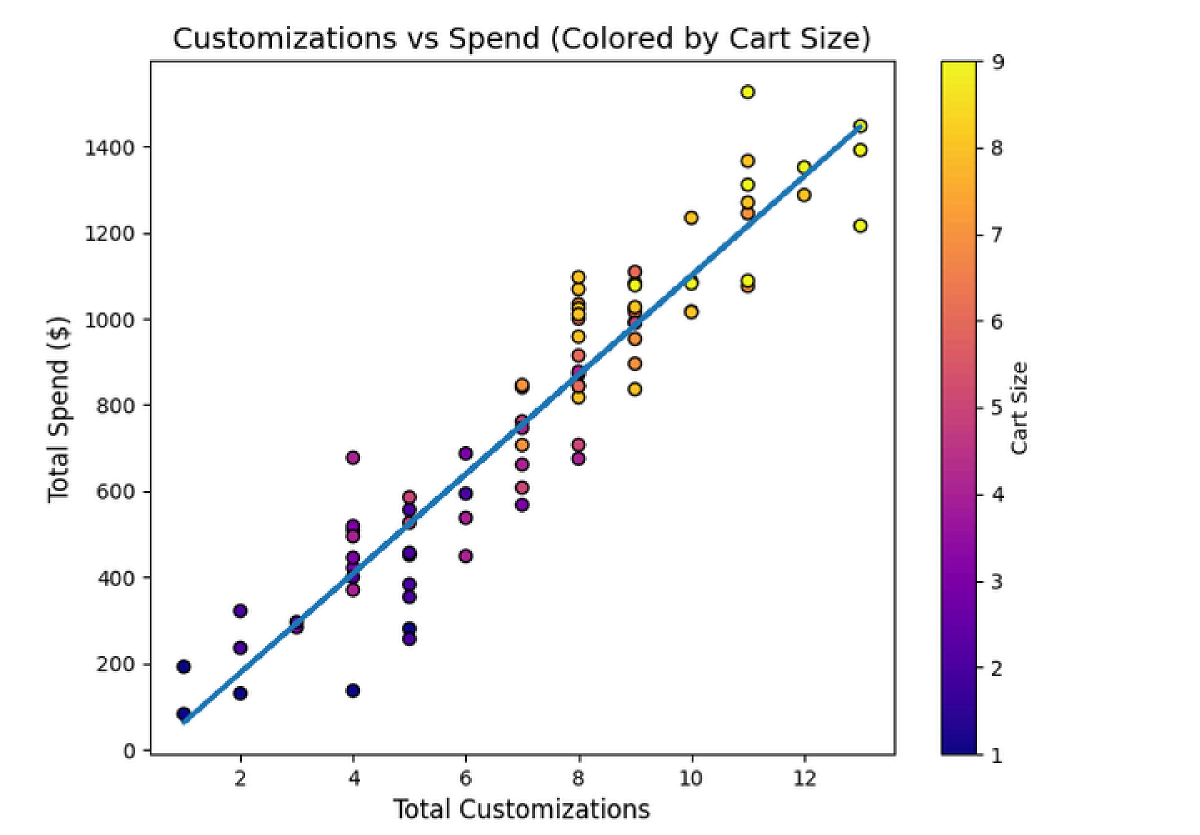
COVARIANCE AND PEARSON CORRELATION-HOW STRONGLY DO CART_SIZE AND NUM_CUSTOMIZATIONS RELATE TO TOTAL_SPEND?





Feature	Habit Engine (Customizations)	Volatility Risks (Cart Size)
Statistical Metric	Pearson Correlation r = 0.95	Pearson Correlation r \approx 0
Relationship Strength	Near-Perfect / Strong	Weak / Non-Linear
Mathematical Logic	Spend increases predictably with every modification (syrups, shots).	Total item count does not reliably predict the final bill.
Covariance (Direction)	Positive (+): Confirms high-margin growth.	Positive (+): Confirms volume adds revenue, but inconsistently.
Visual Proof	Tight, linear cluster of "Routine Commuters."	Scattered "cloud" of dots (One-off bulk buyers).
Strategic Takeaway	Priority 1: Use "Frequency Nudges" and premium upsells.	Risk: Bulk buyers are unforecastable; avoid volume-only discounts.

Business Insight-"Revenue growth is driven more effectively by upselling premium modifications (customizations) rather than just encouraging customers to buy more items per visit (cart size)."

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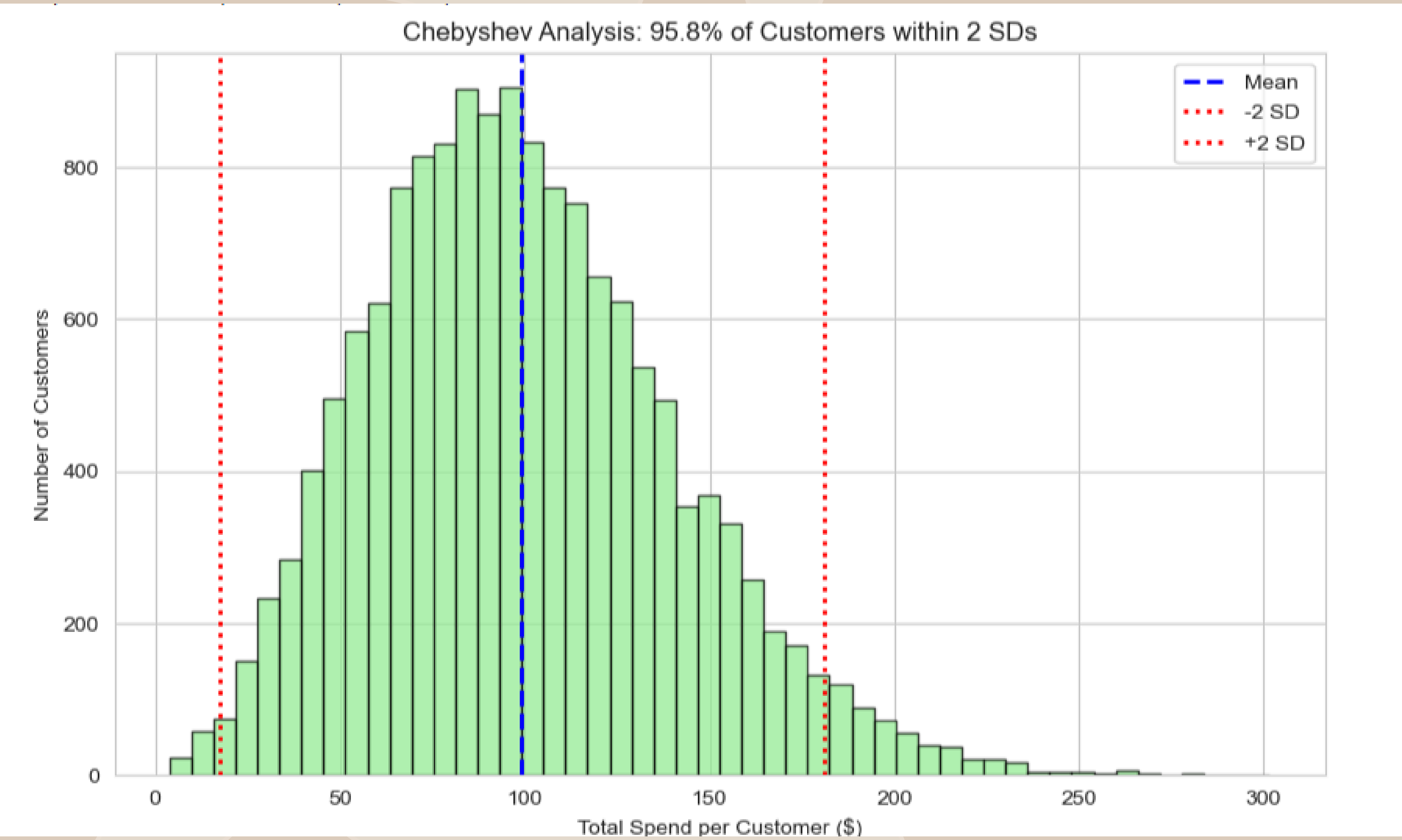
CHEBYSHEV'S INEQUALITY VS EMPIRICAL RULE

"Chebyshev provides a mathematical certainty;
The Empirical Rule provides a precise estimate."

ANALYSIS 2-CHEBYSHEV'S INEQUALITY-WHAT % OF SPENDING FALLS WITHIN 2 SDS? COMPARE WITH EMPIRICAL RULE



Rule/Concept	Expected	Actual (Our Data)	Business Takeaway
Chebyshev	⩾75%	95.80%	Safety: Guaranteed stability.
Empirical Rule	≈95%	95.80%	Stability: High-precision forecasting.
Normality	Symmetric	Right-Skewed	Strategy: Avoid "average" marketing.

Business Insight-"We have achieved a 'High-Confidence' Revenue Model. Even though the math (Chebyshev) only requires 75% stability, our actual customer data hits 95.8%. This beats the standard 95% Empirical Rule and proves that our financial future is not at the mercy of a few 'Whales' or outliers. Our core business is steady, predictable, and ready to scale."

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PROBABILITY & BAYES' THEOREM

Foundations and Spending Thresholds

Random Variable

Total spend is a continuous random variable. It can take any value within a range based on customer behavior.

Event & P(A)

An event is a specific outcome. P(A) is the probability of that event, always between 0 and 1. Here, A means a customer spends more than $25.

P(A | B)

Conditional probability is the probability of A given that B has already occurred. Example: P(high spend | Mobile App user)

Analysis 1 — Spending Threshold Probabilities

Business angle: At what spending level does a customer become genuinely high value, and how many of them exist?

$$Z = (X - \mu) / \sigma$$

μ = $14.87 σ = $5.51 | Zero nulls, 100K rows

Threshold	Z Score	Probability	Est. Customers
Spend > $15	0.024	49%	~49,000
Spend > $20	0.93	17.6%	~17,600
Spend > $25 ★	**1.84**	**3.3%**	**~3,300**

★ At $25 spending becomes genuinely rare. Only 1 in 30 customers cross this line, making it the right threshold for identifying Starbucks highest value customers.

ANALYSIS 2 & 3 — CONDITIONAL PROBABILITY AND BAYES' THEOREM

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Analysis 2 — Who is most likely to be a high spender?

Mobile App	Non Mobile	Rewards Member	Non Member
Mean: \$18.08	Mean: \$12.49	Mean: \$15.72	Mean: \$14.09
SD: \$5.48	SD: \$4.15	SD: \$5.67	SD: \$5.23
Z = 1.26	Z = 3.01	Z = 1.64	Z = 2.09
P(>\$25) = 10.4%	**P(>\$25) = 0.13%**	**P(>\$25) = 5.0%**	**P(>\$25) = 1.8%**

★ **Mobile App users are ~79x more likely to be high spenders than non mobile users (10.4% vs 0.13%), verified via Google Colab**

Analysis 3 — Bayes' Theorem: When a high value order comes in, who placed it?

P(B|A) = P(A|B) × P(B) / P(A) → P(Mobile | high spender) = P(high spender | Mobile) × P(Mobile) / P(high spender overall)

Mobile App

P(high spender | Mobile) = 0.1038
P(Mobile) = 42,521 / 100,000 = 0.4252
P(overall) = (0.1038×0.4252)+(0.0013×0.5748) = 0.04488
= 0.04413 / 0.04488

Result: **98.3%**

Rewards Member

P(high spender | Rewards) = 0.0505
P(Rewards) = 47,717 / 100,000 = 0.4772
P(overall) = (0.0505×0.4772)+(0.0183×0.5228) = 0.03368
= 0.02410 / 0.03368

Result: **71.4%**

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★ KEY FINDINGS & CONCLUSION

3.3%	10.4%	0.13%	98.3%	71.4%
of all customers spend > $25	Mobile App users are high spenders	Non Mobile users are high spenders	chance high spender used Mobile App	chance high spender is Rewards Member

Conclusion

The Mobile App is the single strongest predictor of high value customer behavior at Starbucks. Mobile App users are ~79x more likely to spend above $25, and 98.3% of all high value orders originate from the Mobile App. The app does not just change how customers order. It fundamentally changes how much they spend. All numbers verified via Google Colab against the raw Kaggle dataset (100,000 rows, zero null values).

HYPOTHESIS TESTING

A statistical method used to determine whether observed differences in data are statistically significant or due to random chance.

Aids in making informed, data-driven decisions.

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QUESTION

Do customers spend differently based on:
Order Channel - (Mobile vs Non - Mobile)?
Rewards Membership - (Yes vs No)?

WHY ARE WE USING IT?

To validate if observed differences in spending are statistically significant
Not just based on averages, but mathematically proven

DEFINE HYPOTHESES

Mobile vs Non-Mobile And Rewards vs Non-Rewards
$H_0: \mu_1 = \mu_2$
$H_a: \mu_1 \neq \mu_2$

TEST USED:

Independent Two-Sample t-test
$\alpha = 0.05$

RESULTS & INTERPRETATION

Mobile vs Non-Mobile
Mean: 18.08 vs 12.49
$t = 175.91$
p-value < 0.0001

Reject H_0
(Null Hypothesis)

Mobile users spend more

Rewards vs Non-Rewards
Mean: 15.72 vs 14.09
$t = 47.08$
p-value < 0.0001

Reject H_0
(Null Hypothesis)

Rewards members spend more

Differences are statistically significant

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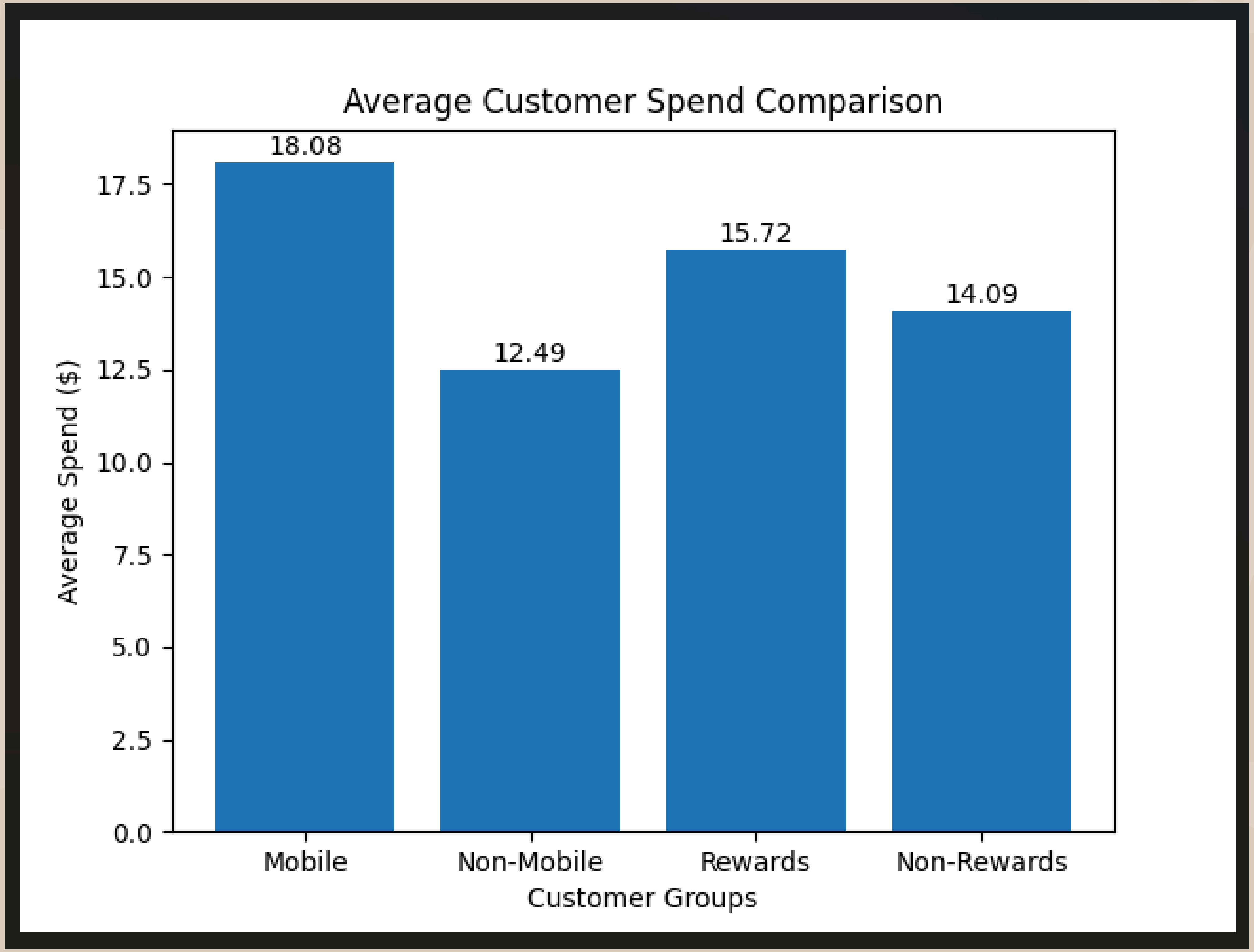
GRAPH REPRESENTATION



NEXT STEPS & REMAINING WORK

What we have completed:

- Data Exploration & Descriptive Statistics (Naman)
- Correlation, Covariance, Chebyshev's inequality Vs Empirical Rule (Priya)
- Probability, Conditional Probability & Bayes Theorem (Varuna)
- Hypothesis Testing: Two-sample t-tests for Mobile vs Non-Mobile and Rewards vs Non-Rewards (Sanjana)

What remains for the final submission (May 5):

- Additional hypothesis tests (paired t-test for repeat customers, two-proportion z-test)
- Confidence intervals for mean spend by segment
- Full academic report in LaTeX (10-12 pages, double-column, J-NaNA format)
- YouTube demo recording (12-15 minutes)
- Deployment discussion: how findings translate to actionable business strategy
- Consolidate all code into final Jupyter notebooks

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"THANKS A LATTE ☕"